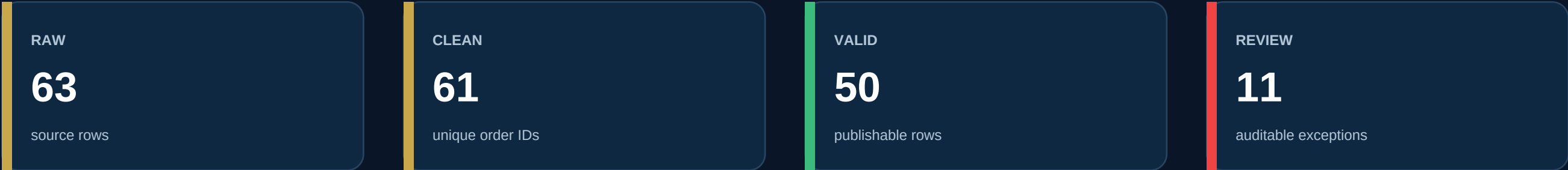


Data Cleaning Before & After

A reproducible retail-order pipeline that standardizes messy operational data, resolves duplicated business keys, calculates trusted sales fields, and separates publishable records from an auditable review queue.



PIPELINE CONTRACT

- Read malformed values as text so the evidence is preserved.
- Apply explicit mappings and parsing rules.
- Prioritize trusted manual adjustments for key conflicts.
- Retain exceptions with row-level issue text.
- Write full, publishable, and review datasets separately.
- Validate uniqueness, schema, counts, and output artifacts.

SOURCE PROFILE

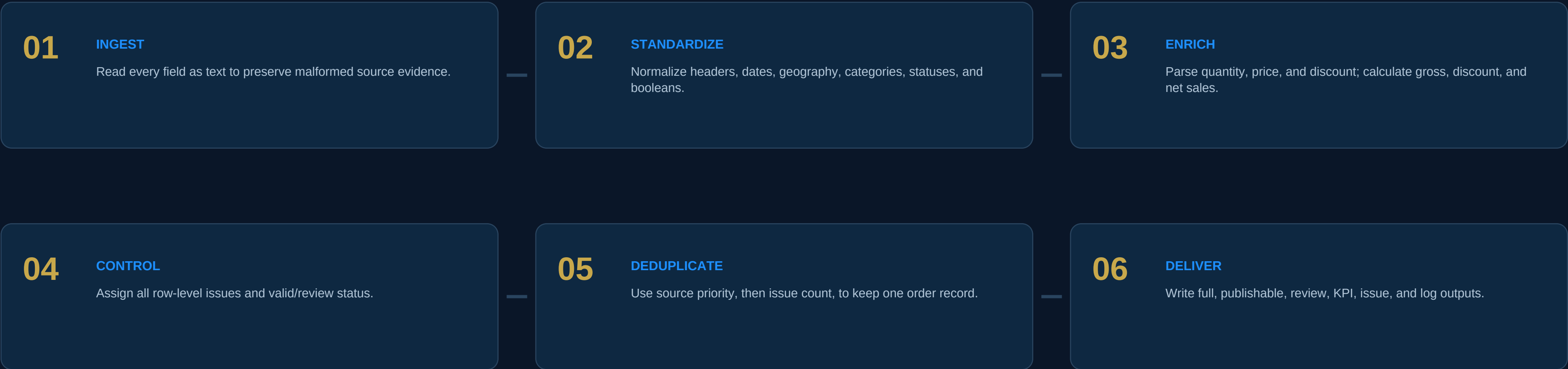
Metric	Value
Rows	63
Columns	22
Logical source files	4
Duplicated order IDs	2
Blank source cells	57

EXAMPLES PRESERVED FOR CLEANING

- Dates arrive as ISO, slash-separated, and ambiguous dash-separated text.
- Money includes symbols, currency labels, commas, blanks, and negative values.
- Country, state, category, payment, shipping, and boolean labels use inconsistent variants.
- Business keys repeat across monthly exports and manual adjustments.
- Missing and invalid records remain visible rather than disappearing during import.

QUALITY CONTROLS TRIGGERED

ambiguous order date format		1
invalid email		1
invalid order date		1
invalid quantity		1
invalid unit price		1
missing category		1
missing customer id		1
missing email		1
missing product sku		1
negative discount		1
negative unit price		1
non positive quantity		1



- ENGINEERING DECISIONS
- Manual adjustments outrank monthly exports for conflicting order IDs.
 - Review-worthy rows remain traceable but are excluded from the publishable output.
 - Built-in unit tests cover parsers, mappings, duplicate priority, status counts, and audit retention.

- Impossible dates stay blank and flagged; the pipeline does not invent a correction.
 - Every transformation can be run from the CLI or demonstrated in the notebook.

Before and After

The output is smaller, typed, standardized, and still auditable

RAW ROWS

63

22 text-heavy source columns

CLEAN ROWS

61

one row per order ID

PUBLISHABLE

50

81.97% valid rate

REVIEW QUEUE

11

18.03% retained for review

BEFORE - SOURCE VALUES

order_id	order_date	country	category	quantity	unit_price	discount
ORD-1002	01/05/2025	United States	Electronics	1	\$18.50	5%
ORD-1007	13/01/2025	USA	Fitness	1	29.99	15%
ORD-1018	2025-01-26	Honduras	Apparel		49.99	0
ORD-1023	02-03-2025	Honduras	Electronics	3	9.99	0
ORD-1002	2025-01-05	USA	Electronics	1	18.50	0.05

Source values retain conflicting duplicates, mixed date syntax, written quantities, currency labels, and inconsistent category or country names.

AFTER - CANONICAL VALUES

order_id	order_date	country	category	quantity	unit_price	discount_rate	quality_status
ORD-1002	2025-01-05	United States	Electronics	1.0	18.5	0.05	valid
ORD-1007	2025-01-13	United States	Fitness	1.0	29.99	0.15	valid
ORD-1018	2025-01-26	Honduras	Apparel		49.99	0.0	review
ORD-1023	2025-02-03	Honduras	Electronics	3.0	9.99	0.0	review

The clean layer provides canonical labels, typed numeric fields, stable dates, one business-key record, and an explicit quality decision for downstream consumers.

Auditability and Delivery

Cleaning is complete only when outputs, exceptions, and contracts are explicit

DELIVERABLES

Artifact	Path	Bytes
clean	cleaned/retail_orders_clean.csv	13925
publishable	cleaned/retail_orders_publishab...	11316
review	cleaned/retail_orders_review.csv	2908
quality_summary	cleaned/data_quality_summary.csv	206
quality_issues	cleaned/data_quality_issues.csv	287
cleaning_log	docs/cleaning-log.md	833

SAFE CONSUMPTION

- Use retail_orders_publishable.csv for analysis that requires valid records only.
- Use retail_orders_review.csv as the remediation queue for data owners.
- Use retail_orders_clean.csv when lineage and all retained business events matter.
- Use quality summaries for monitoring and release evidence.

QUALITY GATES

SCHEMA	Required source columns must exist before transformation.
UNIQUENESS	The final full output must contain one row per order ID.
ROW COUNTS	Clean rows cannot exceed raw rows.
STATUSES	Only valid and review are permitted quality outcomes.
SEPARATION	Publishable and review outputs must be mutually exclusive.
ARTIFACTS	Every declared output must exist after a write-enabled run.

PORTFOLIO SIGNAL

The project demonstrates more than cleanup syntax: it shows source preservation, deterministic rules, exception management, release outputs, and automated contracts - the same controls expected in production analytics pipelines.